

Samuel Tanaka Sibarani

Applying for: AI/ML Security Intern

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Professional Summary

Computer Engineering student specializing in the intersection of artificial intelligence and cybersecurity. Extensive experience developing ML-powered security systems, including fraud detection platforms and network threat analysis tools. Seeking an AI/ML security internship to advance research in intelligent threat detection and automated security systems.

Education

Universitas Indonesia Jakarta, ID
B.Eng. in Computer Engineering 09/2022 - 09/2026
AI/ML Focus: Machine Learning, Deep Learning, Computer Vision, Natural Language Processing

AI/ML Security Projects

Banking Security Platform - AI Fraud Detection ML Security Engineer | 2024

- Developed real-time fraud detection system using advanced Isolation Forest algorithm
- Implemented adaptive ML models with 95%+ accuracy for financial transaction analysis
- Created comprehensive risk assessment engine with customizable detection rules
- Built automated reporting system with statistical analysis and trend visualization

ML Techniques: Anomaly Detection, Feature Engineering, Model Optimization, Real-time Inference

NetGuard ML - Ensemble Threat Detection AI Research Developer | 2024

- Designed ensemble ML framework combining multiple detection algorithms:
 - **Isolation Forest:** Unsupervised anomaly detection for network behavior
 - **LSTM Networks:** Temporal pattern recognition for attack sequence detection
 - **Random Forest:** Feature-based classification for threat categorization
 - **DBSCAN:** Clustering analysis for attack campaign identification
- Simulated advanced persistent threats with ML evasion techniques

Performance: 98% detection rate with <0.5% false positive rate

SpecialistAI System AI Systems Architect | 2025 - Unfinished

- Built modular multi-agent AI system with intelligent task routing and specialist coordination
- Implemented 4-bit quantization for efficient deployment in resource-constrained environments
- Designed role-based AI specialists with security-focused expert models

Technologies: Transformers, Phi-2, PyTorch, Google Colab, Quantization

AI/ML Certifications & Training

AI/ML Professional Certifications 2024-2025

1 **MongoDB AI-Powered Search**

MongoDB University | Vector Search, AI Integration, Machine Learning Applications

2 **MongoDB RAG Applications**

MongoDB University | Retrieval-Augmented Generation, Document Processing, AI Applications

3 **IBM Design Thinking Practitioner**

IBM | AI System Design, User-Centered AI Development, Innovation Processes

Advanced MongoDB Data Science Certifications

- **Advanced Schema Design:** Optimized data structures for ML workloads
- **Schema Design Patterns:** Best practices for AI application databases
- **CRUD Operations:** Efficient data operations for ML pipelines

Technical Skills

Machine Learning Frameworks

- **Deep Learning:** 🧠 TensorFlow, 🔥 PyTorch PyTorch, Keras

- **Classical ML:** 🐼 scikit-learn, XGBoost, LightGBM
- **Data Processing:** 📊 pandas, NumPy, SciPy

AI Security Specializations

- **Anomaly Detection:** Isolation Forest, One-Class SVM, Autoencoders
- **Adversarial ML:** Adversarial attacks, model robustness, defense mechanisms
- **Threat Intelligence:** ML-driven threat hunting and pattern recognition
- **Behavioral Analysis:** User and entity behavior analytics (UEBA)

Programming for AI Security

- **Primary:** 🐍 Python (Intermediate-level)
- **Performance:** 🟢 CUDA, 🟡 OpenCL for GPU acceleration
- **Integration:** 🟠 JavaScript, 🔵 C++ for system optimization

Data & Infrastructure

- **Databases:** MongoDB (Certified), SQLite, PostgreSQL
- **Big Data:** Apache Spark, Hadoop ecosystem
- **Cloud ML:** Experience with distributed training and inference
- **MLOps:** Model versioning, automated pipelines, monitoring systems

Research Interests

AI Security Research Areas

Adversarial Machine Learning

- Model robustness against evasion attacks in security applications
- Defense strategies for ML-based intrusion detection systems
- Privacy-preserving machine learning for sensitive security data

Automated Threat Detection

- Zero-day attack detection using unsupervised learning
- Behavioral analysis for insider threat detection
- Real-time adaptive security systems

AI-Powered Security Analytics

- Intelligent log analysis and correlation
- Automated incident response and threat hunting
- Security orchestration with AI decision-making

Security Foundation

Cybersecurity Knowledge

- **Network Security:** Deep packet inspection, traffic analysis
- **Digital Forensics:** Evidence collection and analysis workflows
- **Incident Response:** SIEM integration and automated response systems

Ethical Hacker - Cisco

- Penetration testing with ML evasion techniques
- Security vulnerability assessment using automated tools

Publications & Contributions

Open Source Contributions

- **NetLab V2:** Open-source cyber range platform with AI-driven attack simulation
- **Security Tools:** Custom ML-based security analysis tools and frameworks

Technical Writing

- CVE analysis with ML-assisted vulnerability research
- Security research documentation at sammtan.github.io/writeups

Languages

Indonesian (Native), **English** (Professional Working Proficiency)

AI Security Portfolio: sammtan.github.io • Research: github.com/sammtan